

Project Report

SMART LIBRARY SEAT OCCUPANCY TRACKER

Project Category: University Based Project/IoT Based

Projexa Team Id- 26E1106- (CODESMITH)

Submitted in partial fulfilment of the requirement of the degree of

BACHELOR OF TECHNOLOGY (CSE(core))

Semester-II
to

K.R Mangalam University

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DECLARATION

We hereby declare that this project report is our original work and has not been submitted elsewhere. The information presented here is true and correct to the best of my knowledge.

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CERTIFICATE

This is to certify that the Project Synopsis entitled, "**SMART LIBRARY SEAT TRACKER**" submitted by "**NIMISH KUMAR** (2501010198), **ANVESHA SAXENA** (2501010144), **AKSHAY KUMAR** (2501010033), **AADIT PRASHAR** (2501010073), **SHIVAM KUMAR** (2501010077) and **VIDHI GOYAL** (2501010083)" to **K.R Mangalam University, Gurugram, India**, is a record of bonafide project work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of the University.

Shahid Ahmad Wani

Date: 20th April 2026

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ABSTRACT

In modern academic institutions, libraries serve as essential learning environments where students engage in study, research, and collaborative activities. However, one of the persistent challenges faced by university libraries is the inefficient utilization of seating resources. During peak hours, students often spend a significant amount of time searching for available seats, which leads to overcrowding in certain areas while other seats remain unused. The absence of a real-time seat monitoring system makes it difficult for both students and library administrators to manage seating effectively.

This research proposes an IoT-based Smart Library Seat Occupancy Tracker, a system designed to monitor and display the real-time availability of library seats. The system uses sensors installed at individual seats to detect whether they are occupied or vacant. The collected data is transmitted to a microcontroller, which processes the information and sends it to a centralized server through wireless communication. The updated seat status is then displayed on a digital dashboard or web interface, allowing users to check seat availability instantly.

The proposed system aims to improve space utilization, reduce the time students spend searching for seats, and enhance the overall efficiency of library operations. Unlike camera-based monitoring systems, the solution ensures user privacy while maintaining accuracy and reliability. Additionally, the system supports the development of smart campus infrastructure by integrating IoT technologies into academic environments. The implementation of this system can significantly improve the library experience for students while providing valuable data insights for library management and resource planning.

Chapter 1

Introduction

1. Background of the project

Libraries are fundamental components of educational institutions, providing students with access to academic resources, study environments, and collaborative spaces. With the increasing number of students enrolled in universities and colleges, the demand for library facilities has grown significantly. One of the major issues faced by students in large libraries is the difficulty in locating available seats, especially during examination periods or peak study hours.

In traditional library environments, seat availability is typically determined through manual observation. Students must physically walk through the library to find vacant seats, which can be time-consuming and inefficient. This process not only wastes valuable study time but also leads to overcrowding in certain sections of the library while other areas remain underutilized. Furthermore, library staff often struggle to monitor seating usage effectively due to the large size and complexity of modern library spaces.

With the rapid advancement of digital technologies, particularly the Internet of Things (IoT), it has become possible to develop intelligent systems that can automate the monitoring and management of physical environments. IoT refers to a network of interconnected devices embedded with sensors, software, and communication technologies that allow them to collect and exchange data in real time. By integrating IoT into library infrastructure, it is possible to create a smart system capable of detecting seat occupancy and providing real-time information to users.

The Smart Library Seat Occupancy Tracker is an IoT-based solution designed to address the issue of inefficient seat utilization in libraries. The system uses sensors installed on each seat to detect occupancy and transmit this information to a microcontroller. The microcontroller processes the sensor data and updates the seat status in a centralized database or cloud server. The processed data is then displayed through a digital dashboard, allowing students to check seat availability before entering the library.

This system not only improves convenience for students but also enables library administrators to monitor seat usage patterns and make data-driven decisions for resource management. By implementing such smart technologies, academic institutions can move toward the concept of smart campuses, where digital systems enhance operational efficiency and user experience.

2. MOTIVATION

The motivation for developing the Smart Library Seat Occupancy Tracker arises from common challenges observed in university libraries. Students often experience difficulty in finding available seats, particularly during busy periods such as examinations, project deadlines, and peak study hours. As a result, they may spend several minutes searching for a place to sit, which reduces productivity and causes frustration.

Another issue is the inefficient utilization of available seating space. In many cases, some seats remain vacant while others are overcrowded because students cannot easily identify which seats are available. Without a proper monitoring system, it becomes difficult for library administrators to manage seating resources effectively.

Existing solutions for monitoring occupancy often rely on camera-based surveillance systems. Although these systems can provide accurate information, they raise concerns regarding privacy and data security. Additionally, camera systems can be expensive to install and maintain, making them impractical for many institutions.

Manual monitoring methods also have several limitations. Library staff cannot continuously track seat occupancy in real time, especially in large libraries with hundreds of seats. Furthermore, manual supervision requires significant effort and does not provide accurate data about seat usage patterns.

The development of an IoT-based seat occupancy tracking system provides a practical and cost-effective solution to these problems. By using sensors and microcontrollers, it becomes possible to automatically detect seat occupancy

and update seat status in real time. This allows students to quickly identify available seats without physically searching the entire library.

The primary motivations behind this project include:

- Improving the efficiency of library seating management
 - Reducing the time students spend searching for available seats
 - Enhancing the overall student experience in library environments
 - Supporting the development of smart campus infrastructure
 - Providing a low-cost, scalable, and privacy-friendly monitoring solution
- By addressing these challenges, the Smart Library Seat Occupancy Tracker aims to create a more efficient, user-friendly, and technologically advanced library environment.

Chapter 2

LITERATURE REVIEW

1. Review of existing literature

ULTRASONIC SENSOR AS AN INVESTIGATIVE TOOL

An ultrasonic sensor is an electronic device that is used to measure distance by using ultrasonic sound waves. It works on the principle of echo reflection, where the sensor sends high-frequency sound waves and calculates the time taken for the waves to return after hitting an object. As an investigative tool in this project, the ultrasonic sensor is used to detect the presence or absence of a person on a seat. It continuously monitors the distance between the sensor and the surface (or user). • If the distance is short, it indicates that the seat is occupied. • If the distance is large, it indicates that the seat is vacant. This helps in collecting accurate, real-time data about seat availability without any manual effort. The ultrasonic sensor is preferred because: • It is low-cost and easy to use • It provides reliable distance measurement • It works without physical contact • It ensures privacy, as it does not capture images or videos

SMART LIBRARY MANAGEMENT USING IoT

Traditional library systems do not provide real-time information about seat availability, leading to time wastage and inefficiency. To address this, IoT-based smart library systems have been proposed. These systems use sensors

and network connectivity to monitor and manage resources efficiently. In such systems, data from sensors is transmitted to a central platform, where it is processed and displayed to users. This allows students to quickly find available seats and reduces overcrowding. It also helps administrators in monitoring space usage patterns. Overall, the system improves space utilization, reduces manual effort, and enhances user experience.

REAL-TIME MONITORING SYSTEMS USING EMBEDDED TECHNOLOGY

Embedded systems are widely used for real-time monitoring applications. These systems consist of sensors, microcontrollers, and communication modules that work together to collect and process data efficiently. In this project, the microcontroller receives data from sensors and processes it to determine seat status. The processed data is then transmitted to a central system for storage and display. These systems are known for their fast response, low power consumption, and reliability. This ensures continuous monitoring and makes the system efficient for real-time applications.

WIRELESS COMMUNICATION IN SMART SYSTEM

Wireless technologies such as Wi-Fi and Bluetooth are commonly used in smart systems for data transmission. These technologies enable seamless communication between hardware components and central systems without need.

In this project, wireless communication helps in transmitting seat occupancy data from the microcontroller to the database or user interface. This ensures real-time updates and reduces installation complexity. It also makes the

system flexible and scalable, allowing easy expansion to larger areas or multiple floors.

REAL-TIME DATA DISPLAY AND USER INTERFACE SYSTEM

User interfaces play a crucial role in making systems user-friendly and accessible. Modern systems use web or mobile applications to display data in a clear and understandable format.

In this project, the processed data is presented through a user interface that shows seat availability in real time. The interface is designed to be simple and easy to use, allowing users to quickly identify vacant seats. A well-designed interface improves user interaction and overall system effectiveness.

CONCLUSION:-

From the above studies, it is observed that real-time monitoring systems using sensors, embedded systems, and wireless communication are widely used for improving efficiency. However, many existing systems are either complex, expensive, or require high maintenance.

This project focuses on providing a simple, cost-effective, and efficient solution for real-time seat occupancy tracking in libraries, ensuring better space management and enhanced user convenience.

Table 1. LITERATURE REVIEW/COMPARITIVE WORK

Project Title	Objectives	Technologies Used	Outcomes and Findings
Smart Library Seat Occupancy System using Pressure & RFID Sensors	To monitor seat occupancy and identify users in library	Pressure Sensors (FSR), RFID, WiFi microcontroller, Web App	Real-time seat tracking system developed, High accuracy (~99%) and reduces seat hoarding
IoT-Based Smart Library Seat Reservation System	To analyze smart seat occupancy and reservation systems	Ultrasonic Sensors, Raspberry Pi, NodeMCU, Cloud (ThingSpeak)	Comparison of different IoT models, Multi-sensor systems improve accuracy and efficiency
Library Occupancy Detection using Computer Vision	To detect occupied seats using AI-based image processing	Camera, Deep Learning (Faster R-CNN), Web App	High accuracy detection system, AI improves automation but increases complexity
Smart Parking System using ESP32 and IR Sensors	detect occupancy and provide real-time availability	To detect availability using distance measurement	Real-time monitoring system, Similar concept applicable to seat tracking
Smart Parking System using Ultrasonic Sensors	detect availability using distance measurement	To detect availability using distance measurement	Efficient occupancy detection, Low-cost, reliable, and easy to implement

2. GAP ANALYSIS

The study of existing systems highlights that various technologies such as IoT, RFID, Computer Vision, and sensor-based approaches have been widely used for occupancy detection and monitoring. These systems are effective in their respective domains and provide high accuracy and automation. However, they are often designed for broader applications like surveillance, parking management, or security systems rather than specifically addressing the problem of seat availability in libraries.

Many existing solutions rely on multiple sensors or advanced technologies such as machine learning and deep learning. While these approaches improve accuracy, they also increase system complexity, cost, and maintenance requirements. For example, RFID-based systems require additional hardware and user identification processes, while computer vision-based systems raise privacy concerns and require high computational power.

In addition, IoT-based reservation systems depend heavily on cloud platforms and network connectivity, which may introduce delays or reliability issues. Some systems also lack a simple and user-friendly interface for real-time interaction, making them less practical for everyday use in libraries.

The proposed Smart Library Seat Occupancy Tracker System addresses these limitations by focusing on a simple, efficient, and purpose-specific design. It uses ultrasonic sensors and a microcontroller to provide real-time seat availability without relying on complex algorithms or expensive hardware. The system ensures privacy, reduces cost, and improves ease of implementation while maintaining reliable performance.

Table 2. GAP ANALYSIS

S.no	EXISTING SYSTEM	LIMITATION GAP	PROPOSED and SYSTEM IMPROVEMENTS
1.	Smart Library with RFID & Pressure Sensors	High cost due to multiple sensors, complex hardware setup, maintenance required	Uses fewer components (ultrasonic sensors), cost-effective and easy to install
2.	IoT-Based Seat Reservation System	Dependence on cloud, complex integration, possible delays	Real-time local processing with faster response
3.	Computer Vision-Based System	Privacy concerns, high computational requirements, expensive cameras	No camera usage, ensures privacy and low processing cost
4.	Smart Parking using ESP32 & IR Sensors	Designed for vehicles, not optimized for human seating behavior	Customized specifically for indoor seat tracking
5.	Ultrasonic-Based Parking System	Lacks user interface and real-time visualization for users	Integrated with user interface for live seat availability display

3. PROBLEM STATEMENT

The primary objective of this project is to design and develop an intelligent, sensor-based system for real-time monitoring of seat occupancy in a library environment. The system aims to utilize ultrasonic sensors to accurately detect the presence or absence of users and transmit this data to a microcontroller for processing. The processed data is then communicated to a centralized system or database, enabling continuous monitoring and real-time updates. Another key objective is to minimize the time and effort required by users to locate vacant seats, thereby enhancing overall efficiency and user convenience.

The project also focuses on optimizing space utilization by providing accurate occupancy information, which helps in reducing overcrowding and ensuring better management of available resources. Additionally, the system is designed to be cost-effective, energy-efficient, and easy to implement, making it suitable for large-scale deployment. Emphasis is also placed on developing a user-friendly interface that presents data in a clear and accessible format. Furthermore, the project aims to build a scalable and flexible architecture that can be extended to other environments such as educational institutions, offices, and smart workspaces.

4. OBJECTIVES

The primary objective of this project is to design and develop an intelligent, sensor-based system for real-time monitoring of seat occupancy in a library environment. The system utilizes ultrasonic sensors to accurately detect the presence or absence of users and transmits this data to a microcontroller for processing. The processed information is then sent to a centralized system or database, enabling continuous monitoring and real-time updates. The project aims to reduce the time and effort required by users to locate vacant seats, thereby improving efficiency and user convenience. It also focuses on optimizing space utilization, minimizing overcrowding, and ensuring better management of resources. Additionally, the system is designed to be cost-effective, energy-efficient, and easy to implement, with a user-friendly interface for clear data visualization. The architecture is scalable and flexible, allowing future expansion to larger environments and integration with smart systems.

Key Objectives:

- To design a real-time seat occupancy detection system using ultrasonic sensors
- To implement efficient data processing using a microcontroller
- To enable real-time data transmission to a central system/database
- To develop a user-friendly interface for displaying seat availability
- To reduce manual effort and time in locating vacant seats
- To improve space utilization and reduce overcrowding
- To ensure low-cost, reliable, and scalable system design

CHAPTER 3

METHODOLOGY

The methodology section of this project defines the systematic approach used to design and implement the Smart Library Seat Occupancy Tracker System. It outlines the procedures, techniques, and technologies applied to achieve accurate and real-time seat monitoring. This section is essential as it provides a clear understanding of how the system functions, how data is collected and processed, and how different components interact with each other. The methodology ensures that the system is efficient, reliable, and capable of delivering the desired results.

3.1 Overall architecture /Flow chart:

The overall architecture of the Smart Library Seat Occupancy Tracker System represents the high-level structure of the system and the interaction between its components. The system consists of sensors, a microcontroller, a communication module, a central database, and a user interface.

Ultrasonic sensors are placed on each seat to detect occupancy by measuring distance. The collected data is sent to the microcontroller, where it is processed to determine whether a seat is occupied or vacant. The processed data is then transmitted through a communication module such as Wi-Fi or Bluetooth to a central system or database.

The database stores and updates the seat status in real time, and the information is displayed to users through a user interface, such as a web or mobile application. This architecture ensures continuous data flow, real-time updates, and efficient system performance.

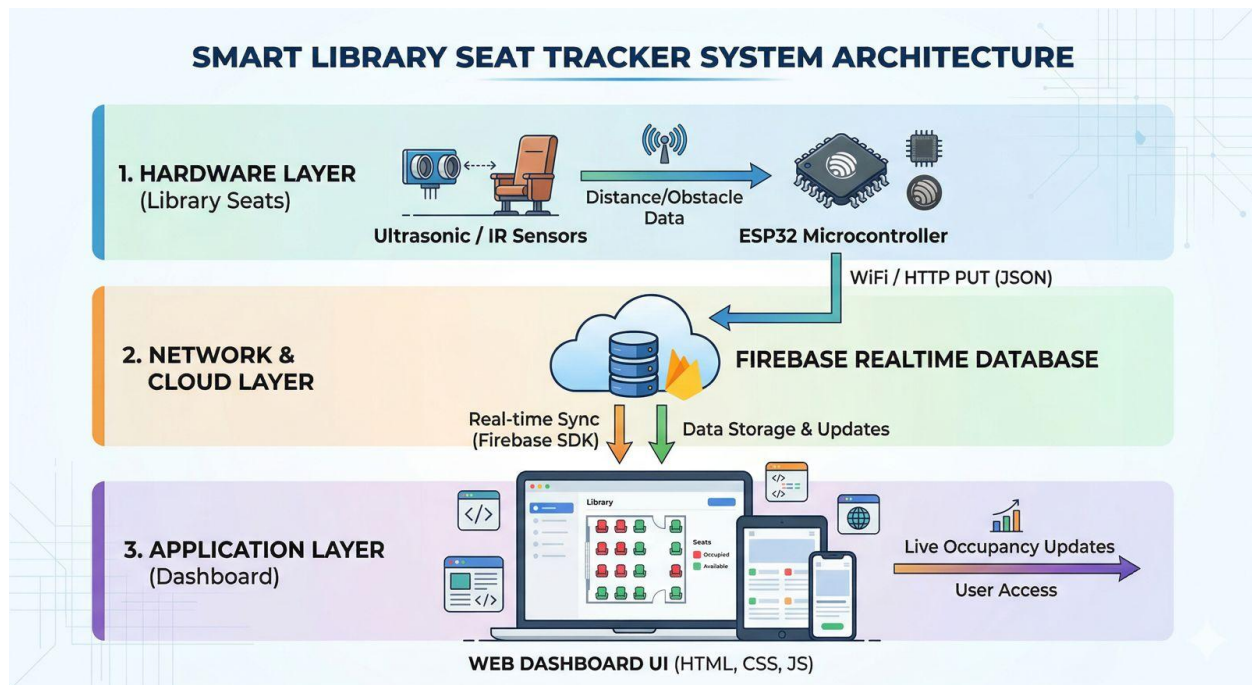


Figure 1. Architecture Diagram

3.2 Data Description

Data Source: The data in this project is generated in real time using ultrasonic sensors connected to an ESP32 microcontroller. These sensors continuously measure the distance between the seat and the user to determine occupancy. The generated data is transmitted to a cloud-based database (Firebase Realtime Database), making the system dynamic and continuously updated.

.Data Collection Process: The data collection process is carried out through sensor-based monitoring. Ultrasonic sensors are placed on each seat to detect occupancy. The sensors emit sound waves and measure the time taken for the echo.

The data collection process follows an automated sensor-based approach:

1. Ultrasonic sensor triggers a pulse and measures distance
2. Distance data is captured by the ESP32 microcontroller

3. The system checks if distance is below a threshold (e.g., < 20 cm)
4. Seat status is determined (Occupied / Available)
5. Data is converted into JSON format
6. Data is sent to the cloud (Firebase RTDB) via HTTP request
7. Database updates and logs timestamp
8. UI fetches updated data in real time

Data Type: The system uses a combination of:

- **Numerical Data** – Distance values (in cm)
- **Categorical Data** – Seat status (Occupied / Available)
- **Time-Series Data** – Timestamped updates stored in database

Data Size: The system generates continuous streaming data. Each seat produces multiple readings over time, making the dataset dynamic and continuously growing. The size depends on the number of seats and frequency of updates.

Data Format: Data is structured in **JSON format** for communication and storage. Example:

```
{  
  "seat_id": "A1",  
  "status": "occupied",  
  "distance": 15,  
  "timestamp": "2026-04-09T10:30:00"  
}
```

Data is stored in Firebase Realtime Database in structured nodes (e.g., /seats/A1).

Data Preprocessing: Raw sensor data is processed at the microcontroller level:

- Noise reduction and filtering applied
- Threshold comparison (e.g., < 20 cm = occupied)

- Only state changes are transmitted to reduce redundant data
This ensures efficient and accurate data handling.

Data Sampling: Since the system works in real time, continuous data is used. However, small delays (e.g., 2 seconds) are introduced between readings to optimize performance and avoid excessive data transmission.

Data Quality Assurance: Data accuracy is ensured through:

- Sensor calibration
- Threshold tuning
- State-change detection logic
- Testing under different conditions

Some minor errors may occur due to environmental noise or object interference.

Data Variables: The main variables used in the system include:

- Seat ID – Unique identifier for each seat
- Distance – Measured value from ultrasonic sensor (in cm)
- Seat Status – Occupied or Vacant
- Timestamp – Time of data recording

Data Distribution and Summary Statistics:

Distance values fall into two main ranges:

- **Low distance** → Seat Occupied
- **High distance** → Seat Available

The system uses these ranges to classify seat status efficiently. Data is continuously updated, enabling real-time monitoring and visualization in the user interface.

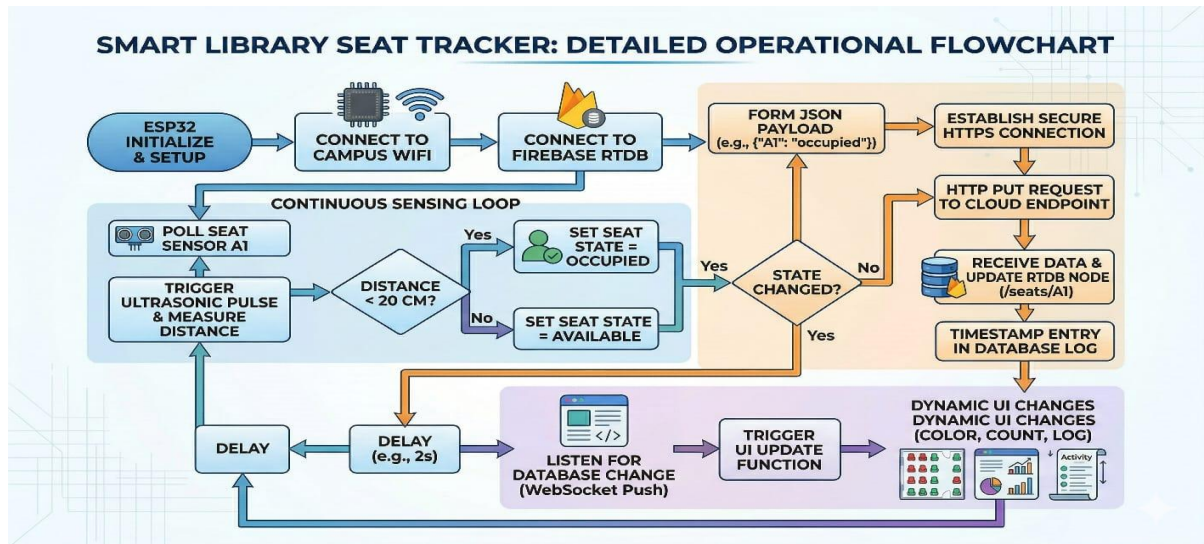


Fig. 2 detailed explanation of diagram

3.3 Implementation

The implementation of the Smart Library Seat Occupancy Tracker System involves the integration of hardware and software components to achieve real-time seat monitoring. The system is developed using ultrasonic sensors, a microcontroller, and a communication module to ensure efficient data collection, processing, and transmission.

Initially, ultrasonic sensors are installed on each seat to detect occupancy by measuring the distance between the sensor and the seat surface. These sensors continuously generate real-time data, which is transmitted to the microcontroller. The microcontroller is programmed to process the incoming data and compare it with predefined threshold values to determine whether a seat is occupied or vacant.

Once the seat status is identified, the processed data is sent to a central system or database using wireless communication technologies such as Wi-Fi or Bluetooth. The database stores and updates the seat information in real time, ensuring that the latest data is always available.

On the software side, a user interface is developed to display seat availability. This interface can be accessed through a web or mobile application, where users can easily view which seats are occupied or vacant. The interface is designed to be simple, responsive, and user-friendly.

The system is tested under different conditions to ensure accuracy and reliability. Calibration of sensors is performed to minimize errors and improve detection performance. The integration of all components ensures smooth data flow and continuous system operation.

Implementation Steps:

- Installation of ultrasonic sensors on seats
- Programming of microcontroller for data processing
- Setting threshold values for occupancy detection
- Establishing wireless communication (Wi-Fi/Bluetooth)
- Developing database for data storage
- Designing user interface for seat display
- Testing and calibration of the complete system

Tools & Technologies Used:

- Ultrasonic Sensors
- Microcontroller (Arduino / NodeMCU)
- Wi-Fi / Bluetooth Module
- Programming Languages (C/C++ / Embedded C)
- Web/App Interface (for display)

3.4 Result and Testing:

The Smart Library Seat Occupancy Tracker System was tested to evaluate its performance, accuracy, and reliability under different conditions. The system successfully detected seat occupancy in real time using ultrasonic sensors and displayed the results through the user interface.

During testing, the sensors were able to accurately measure distance and classify seat status as occupied or vacant based on predefined threshold values. The microcontroller processed the data efficiently, and the communication module ensured smooth transmission of data to the central system. The user interface displayed updated seat availability without noticeable delay, confirming the system's real-time capability. Multiple test cases were conducted by placing and removing objects on seats to simulate occupancy. The system responded quickly to changes, demonstrating consistent performance. Minor variations in sensor readings were observed due to environmental factors, but these were minimized through calibration and threshold adjustments.

Test Cases:

- **Seat Occupied:** Sensor detects short distance → System marks seat as occupied
- **Seat Vacant:** Sensor detects larger distance → System marks seat as vacant
- **Continuous Monitoring:** System updates status dynamically in real time

Results Obtained:

- Accurate detection of seat occupancy
- Real-time data updates with minimal delay
- Stable communication between components

- User-friendly display of seat availability

Performance Analysis:

- High accuracy achieved in occupancy detection
- Fast response time of the system
- Reliable performance under normal conditions
- Minor errors reduced through proper calibration

Details of tools, software, and equipment utilized.

PLATFORM USED

For this project, various modern technologies and components have been used to design and implement the Smart Library Seat Occupancy Tracker System. This section describes the tools, software, and hardware used along with the reasons for their selection.

PROGRAMMING LANGUAGE: EMBEDDED C / ARDUINO IDE

In this project, C++ (using Arduino IDE) is used for programming the microcontroller. Arduino is based on C++ and provides a simple and efficient way to interact with hardware components such as sensors and communication modules.

C++ allows direct hardware control while also supporting object-oriented programming, making it suitable for real-time applications like seat occupancy detection. It ensures fast execution, efficient memory usage, and reliable performance in embedded systems.

Reasons for Selecting this language:

- Efficient for hardware-level programming
- Easy integration with sensors and modules
- Supports Object-Oriented Programming (OOP)
- Fast execution and real-time processing
- Widely used in Arduino and embedded systems
- Simple syntax with strong community support

Key Features:

- Supports real-time data processing
- Object-Oriented Programming (classes, functions, modular code)
- Direct interaction with hardware (GPIO, sensors)
- Low memory and power consumption
- High reliability and performance
- Cross-platform support with Arduino IDE

SYSTEM FEATURES

The Smart Library Seat Occupancy Tracker System includes key features that ensure accurate monitoring and efficient seat management in real time.

1.Seat Monitoring

Sensors continuously monitor each seat to detect occupancy .

2.Accurate Detection

Uses threshold-based logic to determine whether a seat is occupied or vacant.

3.Real-Time Updates

Seat status is updated instantly, allowing users to check availability anytime.

4.Wireless Communication

Data is transmitted using Wi-Fi/Bluetooth, reducing wiring complexity.

5.User-Friendly Interface

Provides a simple interface to easily view seat availability.

6.Low Cost & Scalable

Cost-effective system that can be expanded to larger areas.

7.Privacy Friendly :No use of cameras, ensuring user privacy.

ENVIRONMENTAL SETUP

SOFTWARE REQUIREMENTS

The following software components are required for the development and execution of the system:

- Arduino IDE – Used for writing, compiling, and uploading C++ code to the microcontroller
- C++ (Arduino-based) – Programming language used for system logic and sensor handling
- ESP32 Board Support Package – Required to program ESP32 using Arduino IDE
- Firebase / Cloud Platform – Used for real-time database storage and updates
- Web/Application Interface – For displaying seat availability to users
- Operating System – Windows / Linux / Mac OS (any version supported)

HARDWARE REQUIREMENTS

The system requires the following hardware components:

- ESP32 Microcontroller – Main processing unit
- Ultrasonic Sensors – For detecting seat occupancy
- Wi-Fi Module (built-in ESP32) – For data transmission
- Power Supply – To run the microcontroller and sensors
- Connecting Wires & Breadboard – For circuit connections

PLATFORMS ALREADY TESTED ON:

The system can be tested and implemented on:

- Windows
- Linux

Chapter 4

Implementation

1. Detailed Explanation of Implementation

The Smart Library Seat Occupancy Tracker System is implemented by integrating hardware and software components to achieve real-time monitoring. Ultrasonic sensors are installed on each seat to detect occupancy by measuring the distance between the seat surface and the user.

The ESP32 microcontroller, programmed using C++ in the Arduino IDE, continuously reads sensor data. Based on predefined threshold values (e.g., distance less than 20 cm), the system determines whether a seat is occupied or vacant. Once the status is identified, the data is formatted and transmitted to a cloud-based database (Firebase) using Wi-Fi.

The database stores and updates the seat status in real time. A user interface (web/app) is connected to the database, which displays the updated seat availability to users. The entire process runs in a continuous loop, ensuring real-time updates and smooth system performance.

2. Algorithm / Logic Used

Step-by-step Algorithm:

1. Initialize ESP32 and connect to Wi-Fi
2. Trigger ultrasonic sensor to measure distance
3. Read distance value from sensor
4. Compare distance with threshold (e.g., < 20 cm)
5. If condition is true $\rightarrow$ Seat = Occupied
6. Else $\rightarrow$ Seat = Vacant
7. Check if seat status has changed
8. If changed $\rightarrow$ Send data to Firebase database

9. Update database with timestamp
- 10.Repeat process after a short delay (loop)

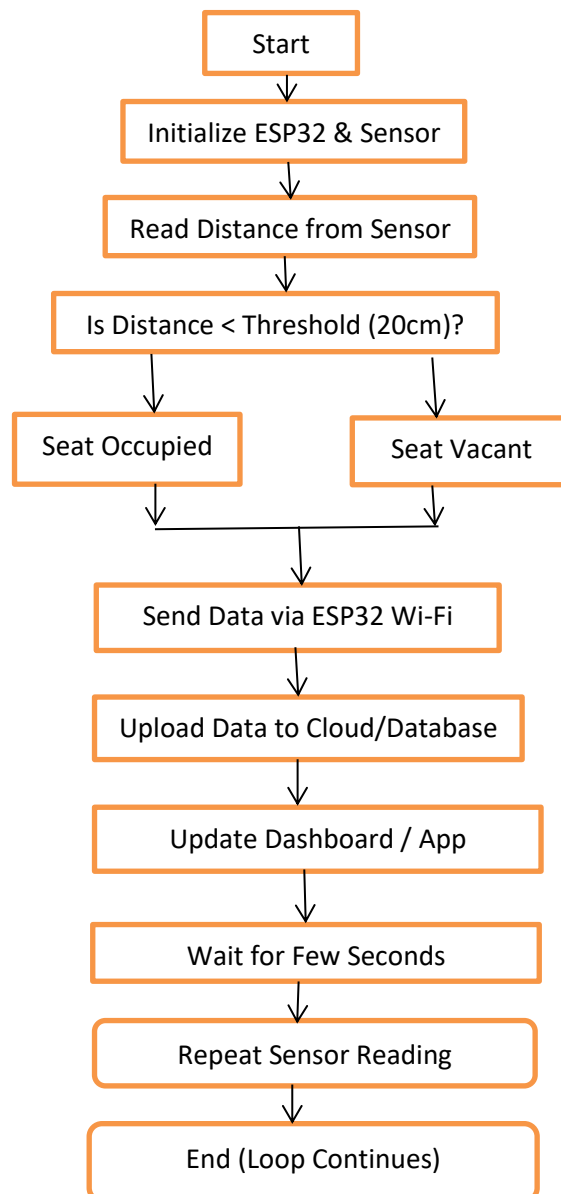
3. Design Overview

The system follows a simple architecture:

Sensor → ESP32 (Processing) → Wi-Fi → Firebase Database → User Interface

This design ensures smooth data flow, real-time updates, and scalability.

4. Flowchart of Work flow of project



5. Challenges Faced and Solutions

Challenge 1: Inaccurate Sensor Readings

- *Problem:* Fluctuations due to environmental noise
- *Solution:* Applied threshold values and filtering techniques

Challenge 2: Delay in Data Transmission

- *Problem:* Slow or inconsistent updates
- *Solution:* Optimized code and reduced unnecessary data transfer

Challenge 3: Continuous Data Overload

- *Problem:* Too much data being sent repeatedly
- *Solution:* Implemented state-change detection (send data only when status changes)

Challenge 4: Connectivity Issues

- *Problem:* Wi-Fi disconnection
- *Solution:* Added reconnection logic in ESP32 code

Chapter 5

RESULTS AND DISCUSSIONS

THE UI Desing of Web page:

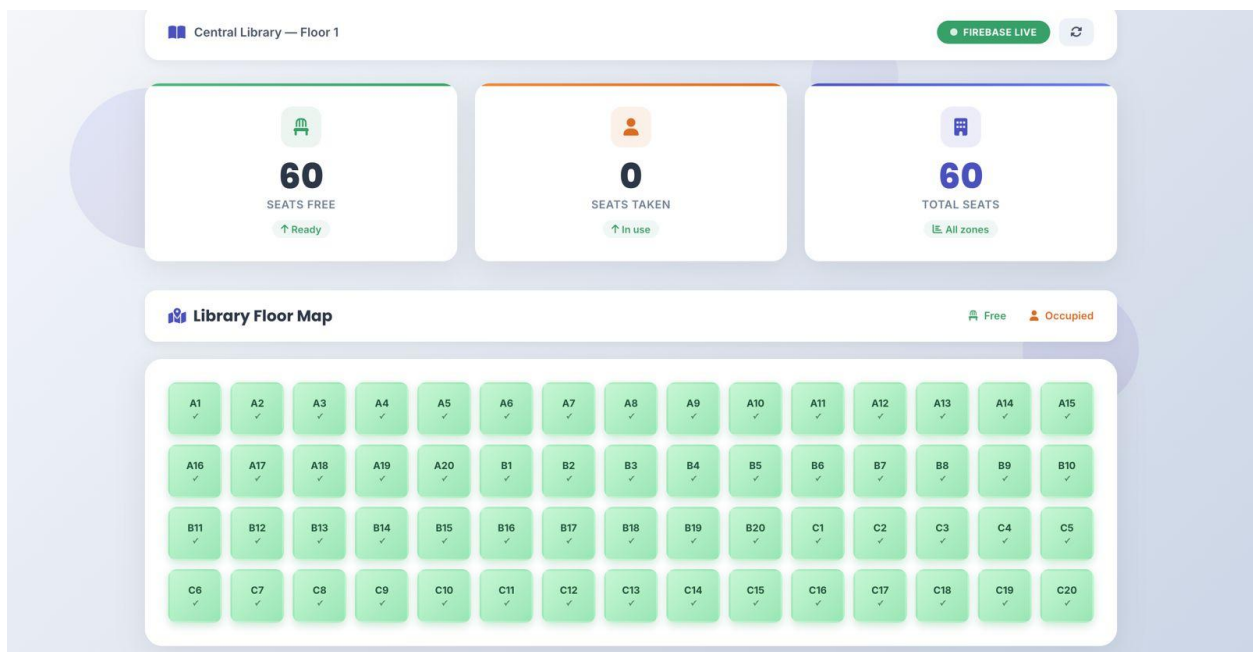
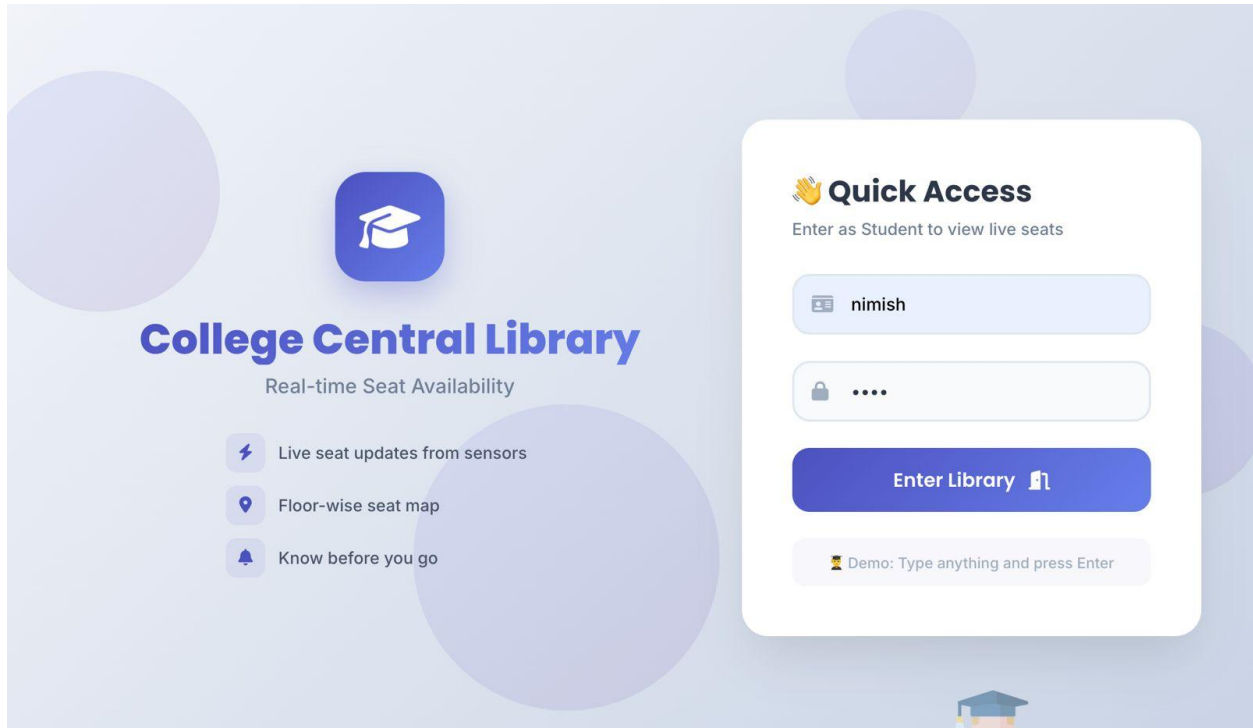
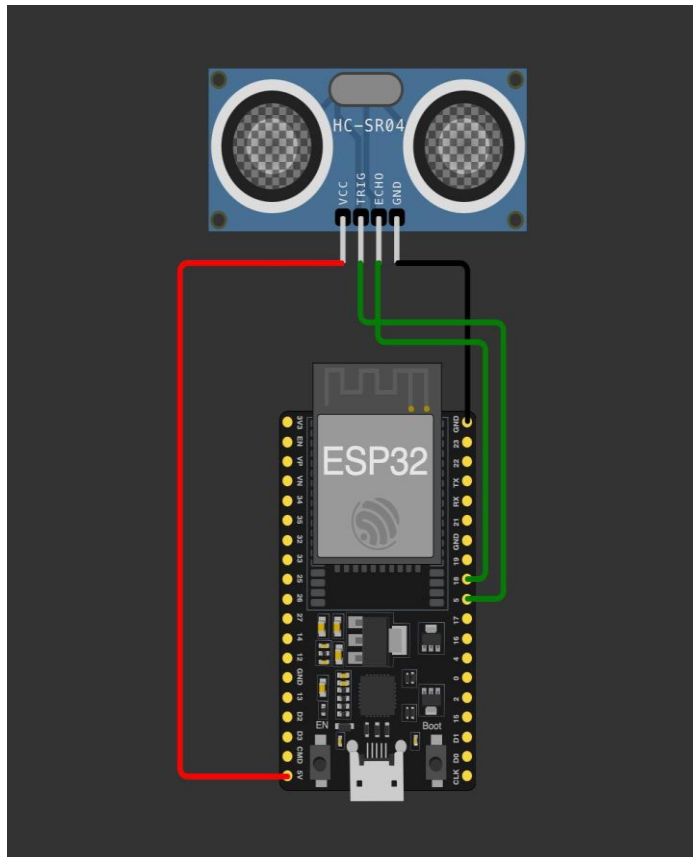
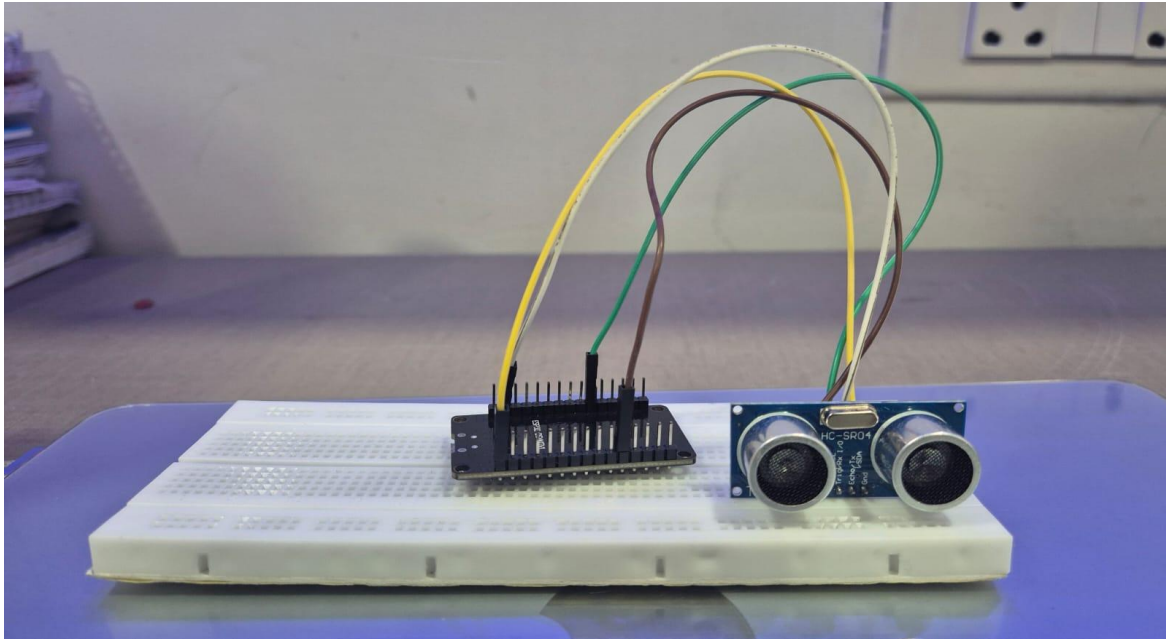


Diagram of IoT device

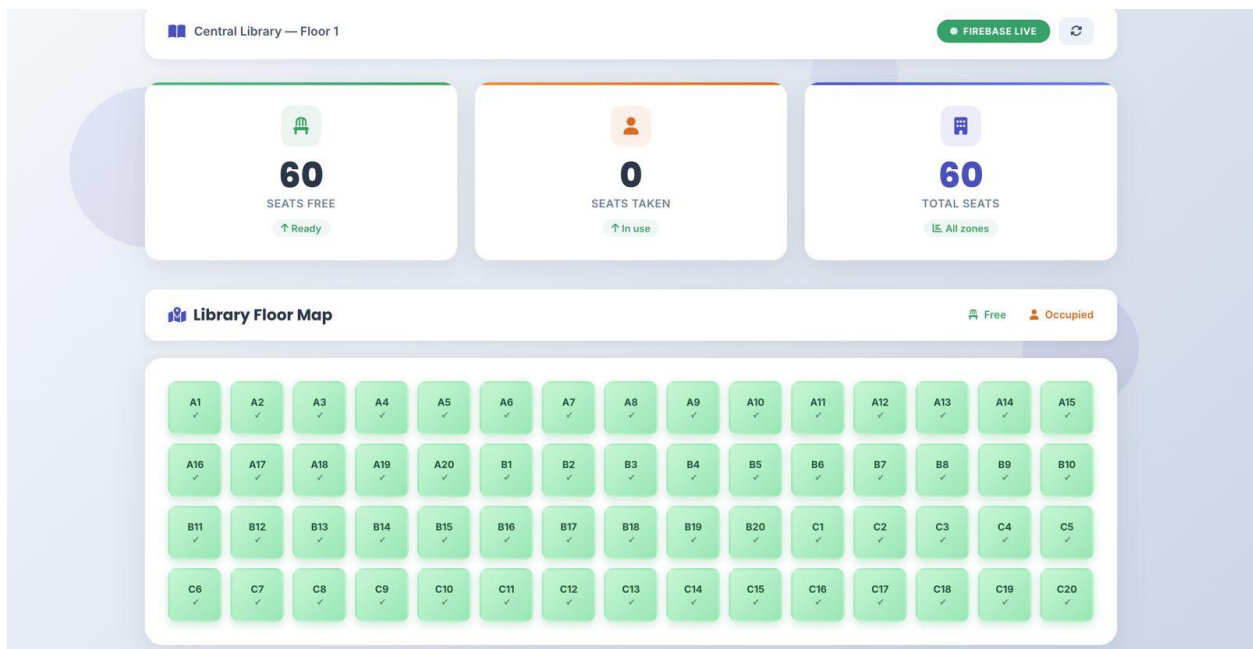


This is tinkercad model of our IoT device. This diagram shows the connection between an **ESP32 microcontroller** and an **HC-SR04 ultrasonic sensor** used for seat occupancy detection. The sensor's **VCC and GND** are connected to the ESP32 for power supply, while the **Trig and Echo pins** are connected to GPIO pins for communication. The ultrasonic sensor measures distance by sending and receiving sound waves, and the ESP32 processes this data to determine whether a seat is occupied or vacant. This setup forms the hardware foundation of the Smart Library Seat Occupancy Tracker System.

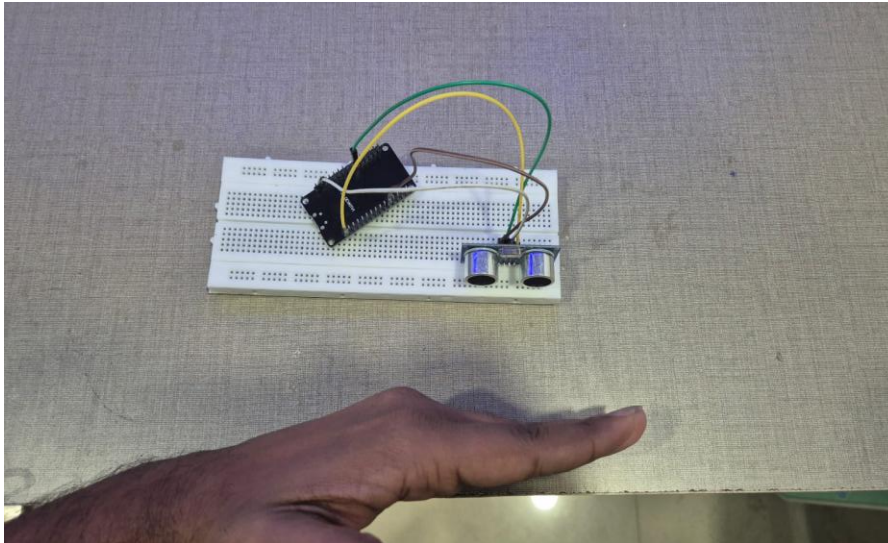
IoT Device:-



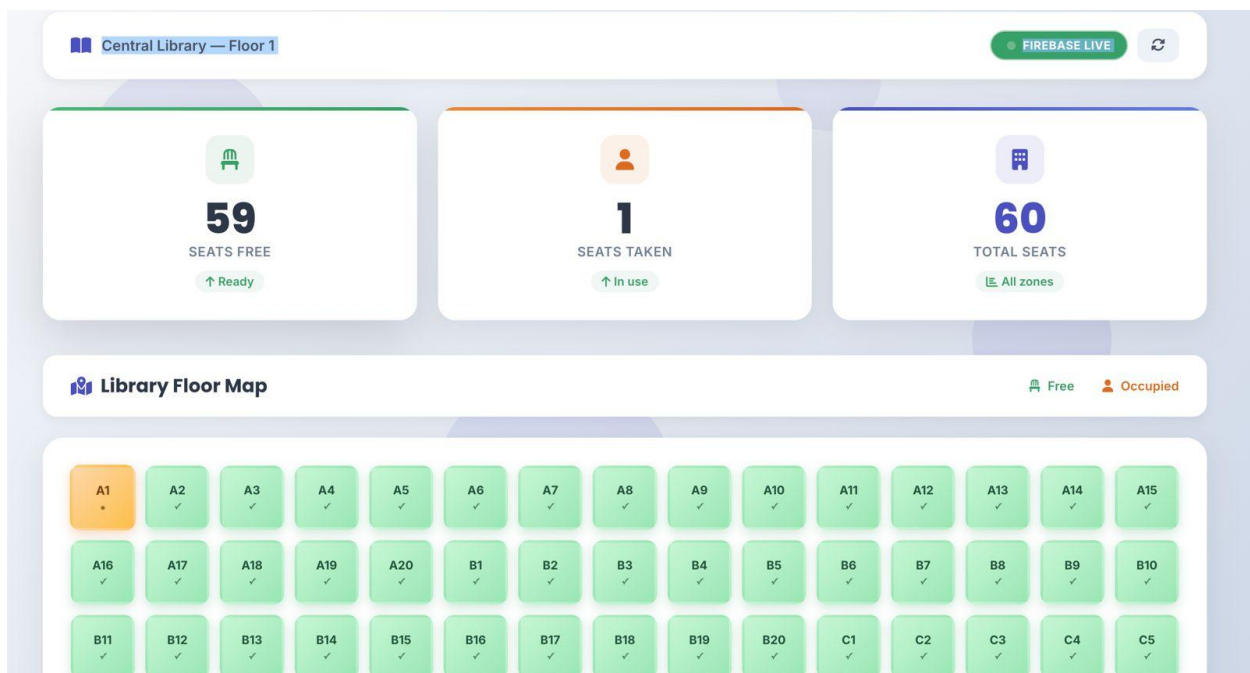
Web Page BEFORE Object Is Placed in Threshold Distance



Object Placed in Threshold Distance:-



Web Page AFTER Object Is Placed in Threshold Distance :-



Chapter 6

FUTURE WORK-

The Smart Library Seat Occupancy Tracker System can be further enhanced by adding advanced features. In future, the system can include a seat booking/reservation system, allowing users to reserve seats in advance through a mobile or web application. This will help reduce conflicts and improve seat management.

The system can also be upgraded using multi-sensor techniques for higher accuracy and reliability. Integration with AI and data analytics can help predict peak usage hours and optimize space utilization. Additionally, features like RFID or user authentication can be added to track user activity and prevent misuse.

Further improvements may include better UI/UX design, cloud optimization, and expansion to larger environments like universities, coworking spaces, and offices, making the system more scalable and efficient.

OUTCOMES AND IMPACTS-

The Smart Library Seat Occupancy Tracker System successfully provides a real-time solution for monitoring seat availability using sensor-based technology. The system is able to accurately detect whether a seat is occupied or vacant and display this information to users instantly, reducing the time spent searching for seats.

Outcomes:

- Real-time seat occupancy detection and display
- Improved efficiency in space utilization
- Reduction in manual monitoring efforts
- Reliable and low-cost implementation using IoT
- User-friendly interface for easy access to information

Impact:

- Saves time for students by quickly locating available seats
- Reduces overcrowding and improves library management
- Enhances user convenience and overall experience
- Promotes smart infrastructure in educational environments
- Provides a scalable solution for universities and public spaces

CONCLUSION-

The Smart Library Seat Occupancy Tracker System provides an efficient and reliable solution to the problem of locating available seats in libraries and similar environments. By integrating ultrasonic sensors with an ESP32 microcontroller programmed in C++, the system is capable of detecting seat occupancy in real time and updating the information through a centralized database and user interface. This eliminates the need for manual searching and significantly reduces time wastage for users.

The project successfully demonstrates the practical application of IoT technology in improving space management and enhancing user convenience. The system ensures accurate detection, fast data processing, and seamless

communication between components. It also maintains user privacy by avoiding the use of cameras or image processing techniques.

Key Achievements:

- Real-time seat occupancy detection and monitoring
- Improved space utilization and reduced overcrowding
- Low-cost and easy-to-implement system design
- Reliable performance with minimal errors after calibration
- User-friendly interface for quick access to seat availability
- Scalable architecture suitable for large environments

The system has been tested under various conditions and has shown consistent and reliable performance. Minor challenges such as sensor fluctuations and connectivity issues were effectively handled through calibration and optimized coding techniques.

Overall, this project highlights the importance of smart systems in modern environments and showcases how simple technologies can be used to solve real-world problems efficiently. With future enhancements such as seat reservation systems, AI-based analytics, and improved user interfaces, the system has strong potential for further development and large-scale implementation in universities, offices, and public spaces.

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ANNEXURE I:

COMPLETE IMPLEMENTATION

The coding part in below with specific modules discussed above.

The database of device is divided into two parts:-

- device.cpp
- firebase.json

Our UI backend is also divided in different some of important parts among them are

- Library floor mapping
- Login handling
- Firebase integration
- Real time updates through live firebase
- Notification or displaying data

device.cpp

This code is used to detect seat occupancy using an ultrasonic sensor and send real-time status updates (occupied/available) to a Firebase database via ESP32 using Wi-Fi.

```
1  #include <WiFi.h>
2  #include <HTTPClient.h>
3  #include <WiFiClientSecure.h>
4
5  #define trigPin 5.
6  #define echoPin 18
7
8  const char* ssid = "nimini";
9  const char* password = "87654321";
10
11 // Firebase URL for Seat A1
12 const char* firebaseURL = "https://library-seat-occupancy-tracker-default-rtdb.firebaseio.com/seats/A1.json";
13
14 long duration;
15 float distance;
16
17 bool lastState = false; // false = available, true = occupied
18 bool firstRun = true;
19
20 void sendToFirebase(String status) {
21   if (WiFi.status() == WL_CONNECTED) {
22     // Firebase uses HTTPS, so you MUST use WiFiClientSecure and bypass the SSL certificate
23     WiFiClientSecure client;
24     client.setInsecure();
25
26     HTTPClient http;
27     http.begin(client, firebaseURL);
28     http.addHeader("Content-Type", "application/json");
29
30     // The REST API expects the data formatted as JSON.
31     String payload = "{" + status + "}";
32
33     int httpStatusCode = http.PUT(payload);
34
35     if (httpStatusCode == 200) {
36       Serial.print("✅ Firebase updated (");
37       Serial.print(status);
38       Serial.println(") HTTP 200 OK");
39     } else {
40       Serial.print("❌ Firebase error: HTTP ");
41       Serial.println(httpStatusCode);
42       if (httpStatusCode == 401 || httpStatusCode == 403) {
43         Serial.println("-> PERMISSION DENIED! Change Firebase Realtime DB Rules to true.");
44       } else if (httpStatusCode == 404) {
45         Serial.println("-> NOT FOUND! Your Firebase Database URL is completely wrong or the database hasn't been created yet!");
46       } else if (httpStatusCode < 0) {
47         Serial.println("-> CONNECTION FAILED! Please check WiFi or SSL.");
48       }
49     }
50     http.end();
51   } else {
52     Serial.println("WiFi not connected!");
53   }
54 }
55
```

```

56 void setup() {
57     Serial.begin(115200);
58
59     pinMode(trigPin, OUTPUT);
60     pinMode(echoPin, INPUT);
61
62     // Connect to WiFi
63     Serial.print("Connecting to WiFi...");
64     WiFi.begin(ssid, password);
65
66     while (WiFi.status() != WL_CONNECTED) {
67         delay(500);
68         Serial.print(".");
69     }
70
71     Serial.println("\nConnected!");
72     Serial.print("ESP32 IP Address: ");
73     Serial.println(WiFi.localIP());
74 }
75
76 void loop() {
77     // Trigger ultrasonic
78     digitalWrite(trigPin, LOW);
79     delayMicroseconds(2);
80
81     digitalWrite(trigPin, HIGH);
82     delayMicroseconds(10);
83     digitalWrite(trigPin, LOW);
84
85     duration = pulseIn(echoPin, HIGH);
86     distance = duration * 0.034 / 2;
87
88     // Determine current seat state
89     bool currentState = (distance > 0 && distance <= 20);
90
91     // Only update Firebase if the state changed, or if it's the first run
92     if (currentState != lastState || firstRun) {
93         if (currentState) {
94             Serial.println("Seat A1 FULL -> Updating Firebase");
95         } else {
96             Serial.println("Seat A1 EMPTY -> Updating Firebase");
97         }
98
99         lastState = currentState;
100         firstRun = false;
101
102         String currentStateString = currentState ? "occupied" : "available";
103         sendToFirebase(currentStateString);
104     }
105
106     delay(500); // Poll sensor twice a second
107 }

```

Firestore.json

This configuration file is used to set up Firebase Hosting by defining the public directory and specifying files/folders to ignore during deployment.

```
1  {
2    "hosting": {
3      "public": ".",
4      "ignore": [
5        "firebase.json",
6        "**/*.*",
7        "**/node_modules/**",
8        "**/*.cpp",
9        "**/*.ino",
10       "**/*.md",
11       "Library_Seat_ESP32/**"
12     ]
13   }
14 }
15
```

App.js

- Library floor mapping

```
1  const CONFIG = {
2    totalSeats: 60,
3    zones: [
4      { name: 'A', label: 'Zone A', count: 20 },
5      { name: 'B', label: 'Zone B', count: 20 },
6      { name: 'C', label: 'Zone C', count: 20 },
7    ],
8    sessionKey: 'lst_student_session',
9  };
10
11  let seats = [];
12  let activityLog = [];
13
14  function initSeats() {
15    seats = [];
16    let id = 1;
17    CONFIG.zones.forEach(zone => {
18      for (let i = 1; i <= zone.count; i++) {
19        seats.push({
20          id: id++,
21          label: zone.name + i,
22          zone: zone.label,
23          zonePrefix: zone.name,
24          status: 'free', // free | busy | booked
25          lastUpdated: new Date(),
26        });
27      }
28    });
29  }
```

- Login handling

```
function handleLogin(e) {
  e.preventDefault();
  const btn = document.getElementById('loginBtn');
  const studentId = document.getElementById('studentId').value.trim();

  if (!studentId) {
    showToast('❌ Please enter your Student ID', 'error');
    return false;
  }

  btn.classList.add('loading');

  setTimeout(() => {
    localStorage.setItem(CONFIG.sessionKey, studentId);
    showToast('✅ Welcome, ' + studentId + '! Loading dashboard', 'success');
    setTimeout(() => {
      showDashboard();
      btn.classList.remove('loading');
    }, 600);
  }, 800);

  return false;
}
```

- Firebase integration

```

208 // ---- Firebase Integration ----
209 function updateSeatFromFirebase(seatLabel, status) {
210   const seat = seats.find(s => s.label === seatLabel);
211   if (!seat) return;
212
213   if (seat.status !== status) {
214     seat.status = status;
215     seat.lastUpdated = new Date();
216
217     activityLog.push({ seat: seat.label, status, time: new Date() });
218
219     const seatEl = document.getElementById(`seat-${seat.id}`);
220     if (seatEl) {
221       seatEl.className = `seat ${status}`;
222       seatEl.title = `${seat.label}  ${capitalize(status)}`;
223       seatEl.innerHTML = `
224         <span>${seat.label}</span>
225         <span class="seat-label">${status === 'free' ? '✓' : status === 'busy' ? '✗' : '🚫'}</span>
226       `;
227       seatEl.classList.add('flash');
228       setTimeout(() => seatEl.classList.remove('flash'), 600);
229     }
230
231     renderStats();
232     renderActivityLog();
233   }
234 }
235
236 function connectToFirebase(firebaseConfig) {
237   if (!firebase.apps.length) {
238     firebase.initializeApp(firebaseConfig);
239   }
240
241   const db = firebase.database();
242   const seatsRef = db.ref('seats');
243
244   const modeLabel = document.getElementById('modeLabel');
245   if (modeLabel) modeLabel.textContent = 'FIREBASE LIVE';

```

- Real time updates

```

247 // Listen for realtime updates
248 seatsRef.on('value', (snapshot) => {
249   const data = snapshot.val();
250   if (data) {
251     // Loop through each seat key (e.g., 'A1', 'A2')
252     for (const [seatLabel, seatData] of Object.entries(data)) {
253       const status = (typeof seatData === 'string') ? seatData : seatData.status;
254
255       let mappedStatus = status;
256       if (status === 'occupied') mappedStatus = 'busy';
257       if (status === 'available') mappedStatus = 'free';
258
259       updateSeatFromFirebase(seatLabel, mappedStatus);
260     }
261   }
262 }, (error) => {
263   console.error('Firebase read failed:', error);
264   showToast('✗ Firebase connection error', 'error');
265 });
266 }
267

```

- Notification or displaying data

```

268 // ---- Toast Notifications ----
269 function showToast(message, type = 'info') {
270   const container = document.getElementById('toastContainer');
271   if (!container) return;
272
273   const toast = document.createElement('div');
274   toast.className = `toast ${type}`;
275   toast.textContent = message;
276   container.appendChild(toast);
277
278   setTimeout(() => {
279     toast.classList.add('toast-exit');
280     setTimeout(() => toast.remove(), 350);
281   }, 3500);
282 }
283
284 // ---- Utilities ----
285 function formatTime(date) {
286   if (!(date instanceof Date)) date = new Date(date);
287   return date.toLocaleTimeString([], { hour: '2-digit', minute: '2-digit', second: '2-digit' });
288 }
289
290 function capitalize(str) {
291   return str.charAt(0).toUpperCase() + str.slice(1);
292 }
293
294 // ---- Boot ----
295 document.addEventListener('DOMContentLoaded', () => {
296   checkSession();
297
298   const firebaseConfig = {
299     apiKey: "58rZCsF1L6BfKrA02H09f7U1mkaum4lkeLLH2ssUQjU",
300     authDomain: "library-seat-occupancy-tracker.firebaseio.com",
301     databaseURL: "https://library-seat-occupancy-tracker-default-rtdb.firebaseio.com",
302     projectId: "library-seat-occupancy-tracker",
303     storageBucket: "library-seat-occupancy-tracker.appspot.com",
304     messagingSenderId: "660542335886",
305     appId: ""
306   };
307
308   if (seats.length === 0) initSeats(); // Ensure seats exist before binding firebase
309   connectToFirebase(firebaseConfig);
310 });

```